NNN-D-001290 (DSA-DM)
December 1, 1969
SUPERSEDING
Fed. Spec. DD-D-205
February 8, 1954

INTERIM FEDERAL SPECIFICATION

DESICCATORS, LABORATORY AND DESICCATOR PLATES

This interim Federal Specification was developed by the Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim Federal specification as a valid exception to the Federal specification DD-D-205.

1. SCOPE

1.1 Scope. This specification covers three types of desiccators and three types of desiccator plates that are suitable for use in general laboratory procedures.

1.2 Classification.

1.2.1 Type, style, grade, and size. Laboratory desiccators covered by this specification shall be of the following types, styles, grades and sizes, as specified: (see 6.2)

Type I - Tubulated cover
Type II - Sleeve-Valve cover
Type III - Plain cover

Style 1 - With three feet (Inside)
Style 2 - Without feet

Grade A - Borosilicate glass
Grade B - Soda-Lime glass

Size 1 - 160 Millimeters (mm) Inside Diameter
Size 2 - 200 Millimeters (mm) Inside Diameter
Size 3 250 Millimeters (mm) Inside Diameter

FSC 6640

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1.2.2 Desiccator plates covered by this specification shall be of the following types, styles and sizes (see 6.2).

Type I - 5 Hole-Plate
Type II - 8 Hole-Plate
Type III - 56 Hole-Plate

Style 1 - With three feet
Style 2 - Without feet

Size 1 - 142 mm Overall Diameter
Size 2 - 230 mm Overall Diameter

2. APPLICABLE DOCUMENTS

2.1 Specification and Standards. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass, Laboratory.
DD-G-675	Ground Joint, Laboratory Glassware, Taper and Spherical.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

STANDARD

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Forth Worth; Denver; San Francisco; Auburn, Washington; and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-L-10547 Liners, Case, And Sheet,Overwrap Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Glass.

3.1.1.1 Grade A. The glass used to fabricate grade A desiccators shall be in accordance with DD-G-541 for Type I Class A or B Borosilicate Glass.

3.1.1.2 Grade B. The glass used to fabricate grade B desiccators shall be in accordance with DD-G-541 for Type II.

3.1.2 Rubber. Rubber components shall be of good quality rubber and shall show no evidence of deterioration when used with stopcock grease, under normal laboratory conditions.

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3.2 Construction. Laboratory desiccators shall consist of a container and cover. When porcelain desiccator plates are required, the type, style and size shall be specified (see 6.2). The contacting surfaces shall be flanged and shall be ground to insure an air-tight fit.

3.2.1 Containers. Style 1 containers shall conform in design to figure 4. The projections on the inside of the container shall serve as feet and shall be equidistant from each other. Style 2 shall be without feet and shall conform in design to figure 5.

3.2.2 Covers.

3.2.2.1 Type I. Type I covers shall be fabricated from Grade A glass and shall be dome shaped with a tubulated knob. The opening in the knot shall accommodate a No. 8 rubber stopper and shall conform in design to figure 1.

3.2.2.2 Type II. Type II covers shall be fabricated from Grade A glass and shall be dome shaped. The dome shall have a knob and shall be fitted with an interchangeable external sleeve with a tooled hose connection at least 25 mm long. Except for the length of the ground zone as specified in Table II, the grinding shall comply with all requirements of DD-G-675. In the open position, the tooled hose connection shall coincide with a round hole in the apex of the cover. The sleeve shall act as a stopcock and, when rotated in a quarter turn from the open position, shall close the desiccator. The open position shall be indicated by an arrow molded into the cap. A rubber ring shall fit into a grooved indentation on the upper area of the cover above the sleeve. The cover shall conform in design to figure 2.

3.2.2.3 Type III. Type III covers shall be fabricated from Grade A or Grade B glass, as specified (see (6.2). The cover shall be dome shaped with a knob on the top. The cover shall conform in design to figure 3.

3.2.3 Desiccator plates. Desiccator plates shall be fabricated from porcelain or other ceramic material and shall be of such quality that it will pass the tests as specified herein:

3.2.3.1 Glaze adhesion. Permanent adhesion of the glaze to another glazed porcelain surface of the same material shall not occur at any temperature below 1150 °C, when tested as specified in 4.4.1.

3.2.3.2 Resistance to chemical attack. Resistance to chemical attack, as measured by loss in weight by boiling in a solution of sodium hydroxide, shall be not more than 0.5 percent, when tested as specified in 4.4.2.

3.2.3.3 Thermal shock. Neither body or glaze of the porcelain shall crack or melt when tested as specified in 4.4.3.

3.2.4 Type I. Type I desiccator plates shall be of a white vitreous, ceramic material. The top surface and edges of the plate shall be glazed and there shall be three evenly spaced feet on the lower surface. The plate shall have 5 holes and shall conform in design to figure 6.

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3.2.5 Type II. Type II desiccator plates shall be fabricated from porcelain and shall be glazed on the edges and top surface. The plate shall have 8 holes and shall conform in design to figure 7.

3.2.6 Type III. Type III desiccator plates shall be fabricated from procelain and shall be glazed on the edges and top surface. The plate shall have one center hole and 56 smaller holes, and shall conform in design to figure 8.

3.3 Pressure resistance. Type I desiccators shall show no evidence of cracking, rupturing, or collapsing when evacuated to 25 mm of mercury when tested as specified in 4.4.4.

3.4 Dimension. All dimensions in this specification are expressed in millimeters (mm).

3.4.1 Containers. The dimensions of Style 1 and Style 2 desiccator containers shall be in accordance with Table I.

Table I. - Dimensions of Style 1 and Style 2 Containers

Size	Inside Diameter	Overall height (Approximate)	Width of flange grinding, minimum
1	145-165	160	13.4
2	195-205	200	15.0
3	245-255	215	17.2

3.4.2 Covers. The dimensions of covers shall be in accordance with Table II.

Table II. - Dimensions of Covers

Size	Inside Diameter	Width of flange grinding, minimum	Standard-Taper Sleeve
1	145-165	20	40/35
2	190-205	25	55/38
3	235-255	27	55/38

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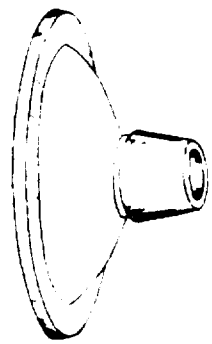


FIG. 1

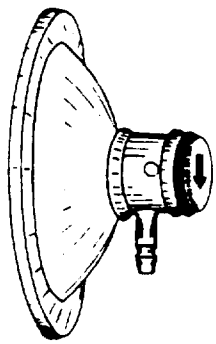


FIG. 2

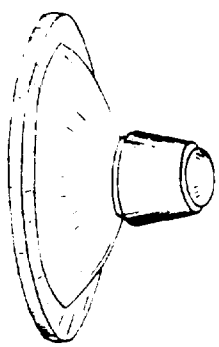


FIG. 3

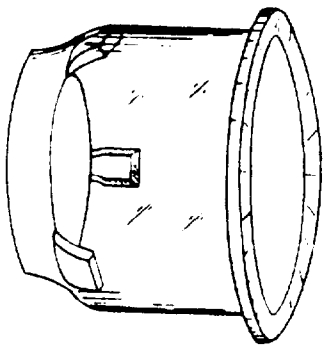


FIG. 4

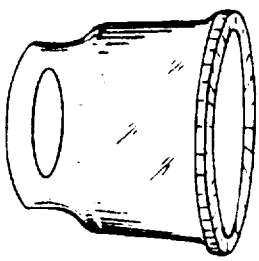


FIG. 5

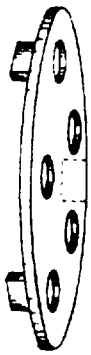


FIG. 6



FIG. 7

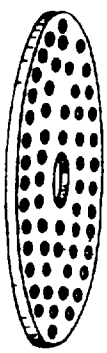


FIG. 8

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3.4.3 Desiccator plates. The dimensions of the desiccator plates shall be in accordance with Table III.

Table III. Dimensions of plates

Type	Overall Diameter	Thickness	Height of Feet	Inside Dia. of Desiccator to accommodate	Hole Diameter
I	142	6	15-16	150-165	30
II	230	6	-----	230-255	30
III	230	6	-----	230-255	5

3.5 Identification mark. Desiccators and plates shall be legibly and permanently marked with the name or registered trade-mark of the manufacturer.

3.6 Workmanship. Workmanship shall be first class throughout. Laboratory desiccators shall be free from defects which detract from their appearance or may impair their serviceability. Laboratory desiccators shall be smoothly finished, symmetrical and practically free of bubbles, blisters, and other imperfections. The thickness of the glass shall be properly and uniformly distributed. Plates shall be free from uneven porcelain distribution.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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4.1.2 Certificate of quality. Certificate of quality, supplied by the manufacturer of the borosilicate glass (see 3.1.1), may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 SAMPLING.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptance quality levels shall be as indicated in Table IV. Unit of product for examination purposes shall be the desiccator.

Table IV. Sampling for Examination

Type defect	Inspection level	AQL (Percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	4.0

4.2.2 Sampling for tests. Unless otherwise specified, sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-4.

4.3 Examination. The desiccator, or the desiccator plate, as specified (6.2) shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with Tables V and VI. The desiccators and desiccator plates shall be examined for defects including, but not limited to, those listed in Tables V and VI.

Table V. Classification of Defects for Desiccators

Categories	Defects
Major	
101	Desiccator not free of cracks.
102	Desiccator not free of chips and breaks.
103	Desiccator not free of open blisters.
104	Desiccator not free of blisters over 3/32 inch.
105	Desiccator not free of stones over 1/32 inch.
106	Desiccator not free of checks, crizzles, and splits.
107	Top surfaces of flange of container not ground.

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Table V. Classification of Defects for Desiccators (Con't)

Major	
108	Top flange not horizontal throughout when the container is placed on a horizontal support.
109	In open position, opening of hose connection does not coincide with hole in apex of cover.
110	Flange on bottom of cover not ground to fit container.
111	Weight of cover not supported by rubber ring, when lifting cover by sleeve.
112	Desiccators not effectively closed when sleeve is rotated a quarter turn.
113	External sleeve does not rotate when vacuum exists inside of desiccator.
Minor	
201	Desiccator not free of clustered bubbles.
202	Desiccator not free of thin areas.
203	Surfaces not free of unspecified protrusions.
204	Desiccator not free of blister, 1/16 to 3/32 inch.
205	Desiccator not free of rough seams or edges, seeds, tears, and deformed areas.
206	Desiccator not free of bloom and laps.
207	Name or registered trademark of the manufacturer not legibly or permanently marked.
208	Desiccator not free of rocking when placed on a smooth flat surface.
209	Projections inside container not positioned in line so as to support plate without wobble.
210	Desiccator not smoothly finished.
211	Desiccator not symmetrical.
212	Projections inside container not equidistant from each other.
213	Hose connections not equidistant from top and bottom of sleeve.

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Table VI. Classification of Defects for Desiccator Plates

Categories	Defects
Major	
101	Desiccator plate not free from cracks, chips, blisters, drizzles and pits
102	Desiccator plate not uniformly glazed on the top surface and edges.
103	Desiccator plate not free from sharp edges and protrusions
104	Desiccator plate not free from rocking or wobbling on a plane flat surface.
Minor	
201	Desiccator plate not free from grease, dirt or objectionable foreign material.
202	Name or registered trademark of the manufacturer not legibly or permanently marked.

4.4 Sampling for Tests. Tests shall be conducted to determine compliance with specification requirements. Unit of product for test purposes shall be as specified in Table VII.

Table VII. Sampling for Tests.

Unit of product	Characteristics	Requirement	Test procedure	Inspection level	AQL
Desiccator plate	Glaze Adhesion	3.2.3.1	4.4.1	S-2	2.5
Desiccator plate	Resistance to chemical attack	3.2.3.2	4.4.2	S-2	2.5
Desiccator plate	Thermal shock	3.2.3.3	4.4.3	S-2	2.5
Desiccator and cover	Pressure Resistance	3.3	4.4.4	S-2	2.5

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4.4.1 Glaze adhesion. Place one plate on top of another so that the glazed surfaces are in contact with each other, and heat both together to a temperature of 1150° C. for a period of 30 minutes, then allow to cool by radiation to room temperature.

4.4.2 Resistance to chemical attack. Each desiccator plate to be tested shall be boiled in 10 percent solution of sodium hydroxide for 8 hours.

4.4.3 Thermal Shock. Plates to be tested shall be heated to a bright red glow (approximately 1000° C) in an electric muffle-type furnace, then removed and allowed to cool by radiation to room temperature.

4.4.4 Pressure resistance. The desiccator, at a temperature of 20° to 25° C., shall be gradually evacuated to 25 mm of mercury and maintained at this pressure for 30 minutes. Air shall then be admitted into the desiccator and a visual examination made for cracks or rupture.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C as specified (see 6.2).

5.1.1 Unit of issue. One desiccator with cover or one cover as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. (Desiccator). The container, cover, plate, and stopcock shall be individually wrapped in kraft paper or other suitable material. Each unit shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Marking shall include the legend: "GLASS - HANDLE WITH CARE"

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend: "GLASS - HANDLE WITH CARE"

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6. NOTES

6.1 Intended use. The desiccators and desiccator plates covered by this specification are intended for use as airtight containers for dehydration, and for keeping materials in an atmosphere of uniform humidity.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, style, grade and size of desiccator when required (see 1.2.1).
(c) Type, style and size of desiccator plate when required (see 1.2.2 and 3.2).
(d) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 Safety precaution. The desiccator shall be enclosed in a heavy, fine mesh wire shield during the evacuation to protect the tester from flying glass in case the desiccator ruptures.

6.4 This specification does not cover all types, classes, styles or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

6.5 Classification changes. The following classification changes have been made:

New designation NNN-D-001290		Old designation DD-D-205	
Type I -	Tubulated cover		
Type II -	Sleeve-Valve cover**	Type I -	Tubulated cover
Type III -	Plain cover	Type II -	Plain cover

** NOTE: The cover identifications used in this specification are classified in accordance with commercial practices. It should be noted that the tubulated cover designated in the old document is referred to as a Sleeve-Valve cover in the new document.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

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